

PSEUDOCODE

- Initialize all devices
- Connect devices to the ISP Router
- Assign IP addresses (Static/Dynamic/Public/Private)
- Verify network connectivity
- Configure DNS server for www.oie.org
- Test DNS resolution

- Register motion IoT devices with IOE server
- While (true):
- For each motion detector:
- If motion detected:
 - o Send alert to IOE server
 - o Trigger CCTV and sirens
 - o Notify user via Laptop:

- Register smoke IoT devices with home gateway
- For each smoke detector:
- If smoke/fire detected:
 - o Send alert to Home gateway
 - o Trigger fire sprinklers and sirens
- Trigger IoT door ,IoT window ,Fire Sprinkler :
 - o Notify user via smartphone
- While (true):
- If user logs in via Laptop ,smartphone respectively:
- Authenticate user
- Grant access to system
- If user views real-time data:
 - o Display data from detectors and webcams
- If user controls devices:
 - o Execute user commands (activate/deactivate alarms, control IoT windows/doors/Fire Sprinkler)
- Periodically:
- Update software
- Perform system health checks
- Backup logs and configuration data